

Problem 9

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Problem 1 Find the derivatives of the following functions:

(a) $f(x) = \sec^{-1}(\sqrt{x})$.

Explanation. $f'(x) = \frac{1}{\sqrt{x}\sqrt{x-1}} \cdot \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2x\sqrt{x-1}}$

(b) $g(x) = \ln(\sin^{-1}(x))$.

Explanation. $g'(x) = \frac{1}{\sin^{-1}(x)} \cdot \frac{1}{\sqrt{1-x^2}}$.

(c) $h(x) = \frac{1}{\tan^{-1}(x^2 + 4)}$.

Explanation. $h'(x) = -(\tan^{-1}(x^2 + 4))^{-2} \cdot \frac{1}{1 + (x^2 + 4)^2} \cdot (2x)$.